# GOOGLE AGENT DEVELOPMENT KIT 2.0 - MASTERCLASS

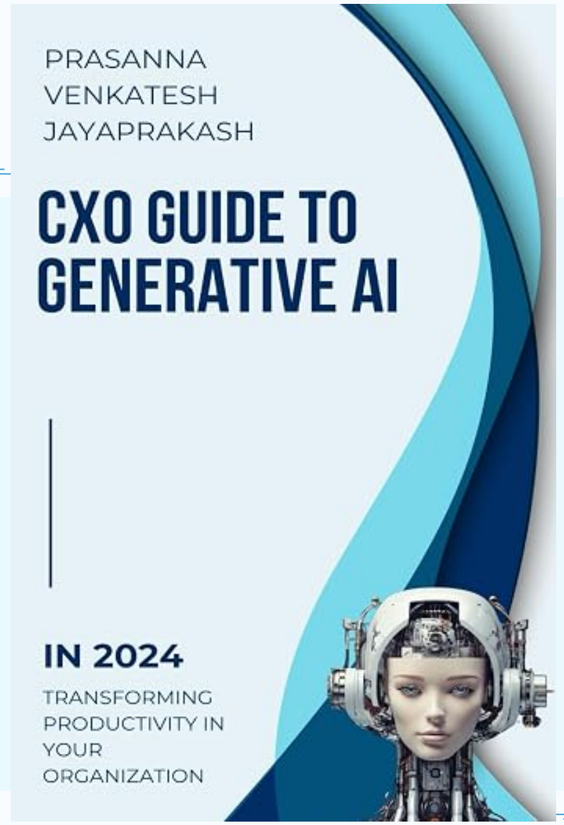
by Dr.Prasanna Jayaprakash

# Agenda

| S.No | Topic | Timing |
|---|---|---|
| | Introduction | 5 Min |
| 1 | Agentic AI - Evolution  & ADK Intro | 10 Min |
| 2 | Live Demo - Simple Agents with ADK | 10 Min |
| 3 | Live demo - Workflow Agents | 10 Min |
| 4 | Live Demo - Human in the loop | 10 Min |
| 5 | Live Demo - Agents with MCP | 20 Min |
| 6 | Local and Global Memory | 5 Min |
| 7 | Q&A | 15 Min |

# Dr.Prasanna - Introduction

- Over 20+ Years in IT
- Worked across US, UK and Middle east
- Currently Leading Global Data science team for system integrator managing all AI, GenAI and Agentic AI programs




- Nvidia Certified Agentic AI Professional.
- On a Mission to Transform 100K companies using AI.

# AI Progression: From Transformers to Agentic AI

Simple explanation with example models



2017

## Transformers

A new AI design that helped computers focus on the most important words in a sentence.

**Examples:**
Transformer, BERT (foundation idea)



2018–2020

## Pretrained Models

Models were trained on huge amounts of text first, then adapted for specific tasks.

**Examples:**
BERT, T5, GPT-2

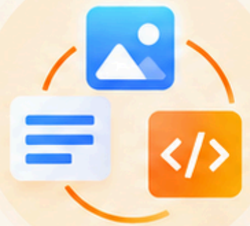


2020–2022

## LLMs

Large Language Models learned to understand and generate human-like text.

**Examples:**
GPT-3, PaLM, LLaMA



2022–2024

## Generative AI

AI started creating useful content like text, images, code, and chat responses.

**Examples:**
ChatGPT, Gemini, Claude, DALL·E



2023+

## Agentic AI

AI moved beyond chatting and can now plan steps, use tools, and complete tasks.

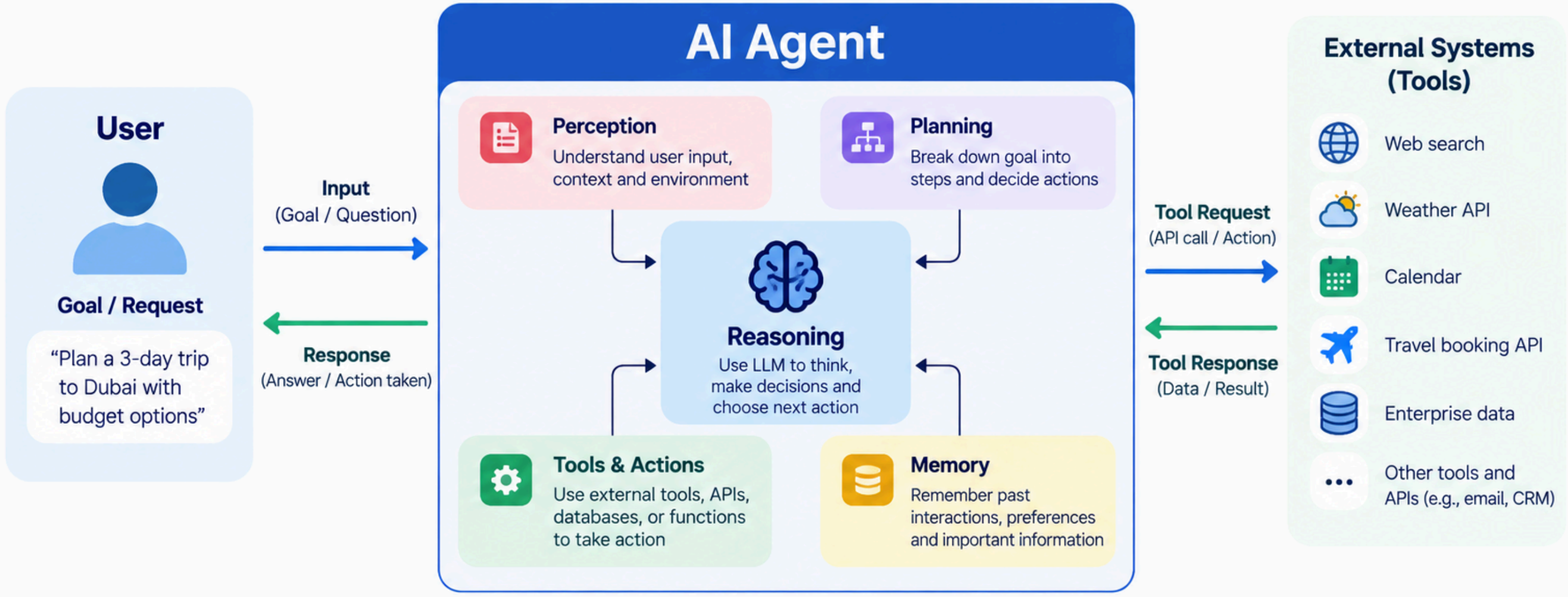
**Examples:**
AutoGPT, Google ADK agents, tool-using copilots



**Attention** →  **Pretraining** →  **Large Language Models** →  **Content Generation** →  **Goal-Driven Agents**

# What is an AI Agent?

An AI agent is an intelligent system that can understand a goal, reason, take actions using tools, and learn from results to achieve the goal with minimal human intervention.




## Example Flow

1. **Understand** — User asks for a travel plan.
2. **Plan** — Decide steps (search, compare, create itinerary).
3. **Take Actions** — Use web search and booking APIs.
4. **Remember** — Save user preferences.
5. **Respond** — Provide a complete travel plan with options.

# GOOGLE ADK EVOLUTION | 1.x → 2.0

From hierarchical agent orchestration to graph-based workflow runtime

## ADK 1.x
### Agent-Centric Foundation

Initial releases (1.x)
2023 – 2025

- **LLM agents & subagents**
  Create agents with system prompts and specialized subagents
- **Tool and function calling**
  Integrate external tools, APIs and custom functions
- **Sequential / Parallel / LoopAgent**
  Orchestrate multi-agent flows using predefined patterns
- **Sessions, state & memory**
  Manage conversation state and memory across interactions
- **Callbacks, evaluation & debugging**
  Hooks for tracing, logging, evaluation and debugging
- **MCP / A2A integrations**
  Connect with Model Context Protocol (MCP) and Agent-to-Agent (A2A)
- **Deployment and observability**
  Run locally or on cloud with basic telemetry and monitoring

**Hierarchical agent orchestration (1.x)**




**Main limitation:**
Predefined hierarchical workflow patterns (sequential, parallel, loop).
Complex branching and dynamic orchestration needs custom implementation.

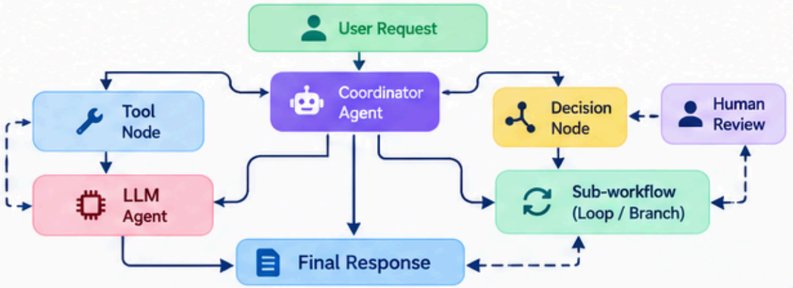
**Architecture shift**
Agents + Tools + Functions become workflow nodes

## ADK 2.0
### Workflow Runtime

GA Release
19 May 2026

- **Graph-based nodes and edges**
  Model workflows as a graph of agents, tools and functions
- **Dynamic branching and loops**
  Flexible control flow based on runtime conditions
- **Fan-out / fan-in & retries**
  Run multiple paths in parallel, merge results and handle retries
- **Collaborative coordinator/subagent flows**
  Advanced multi-agent collaboration patterns
- **Nested workflows & state management**
  Compose reusable sub-workflows with shared state
- **Human-in-the-loop controls**
  Add approval, review and human intervention in the workflow
- **Existing 1.x tools, sessions & agents retained**
  Continue to use 1.x capabilities (with some compatibility caveats)

**Graph-based workflow runtime (2.0)**




**Caution:**
Migration involves changes to agent execution model, event schemas and custom session storage.

1. **ADK 1.x** — Hierarchical agents
2. **ADK 2.0** — Graph runtime
3. **Enterprise workflows** — More control & reliability

**Python ADK 2.0 GA:** 19 May 2026 | ADK 2.0 is not the first version with tools, memory, or multi-agent support.

# Googles Option for developing Agents

01. Python SDK (ADK) & Other Language SDK (TypeScript, Go, Java, or Kotlin)

02. ADK Visual Builder

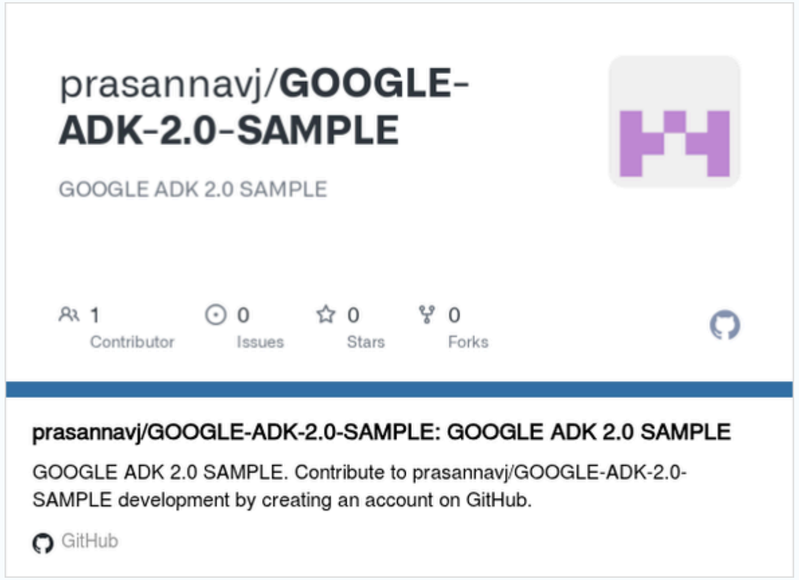
03. Google Agents CLI

04. AI Assisted Development (Antigravity)

05. Managed Agents

# All Sample code available in GitHub

https://github.com/prasannavj/GOOGLE-ADK-2.0-SAMPLE

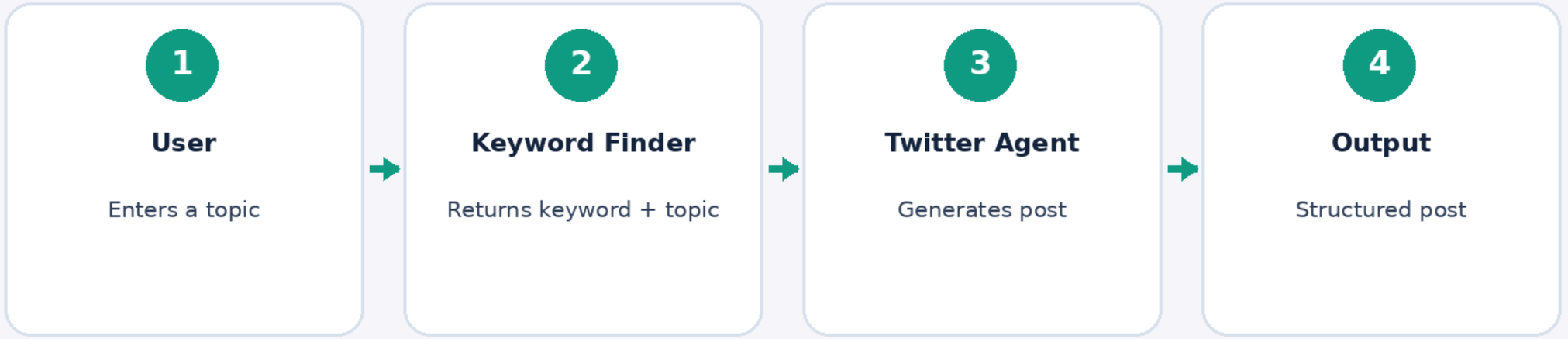


# Workflow Agents

3 workflow_basic

## Sequential Workflow

**OUTPUT FROM AGENT 1 BECOMES INPUT TO AGENT 2**



04 workflow_router

# Conditional Workflow — Cricket Stats

## BRANCH BASED ON THE PLAYER TYPE




### WHAT STUDENTS SHOULD UNDERSTAND

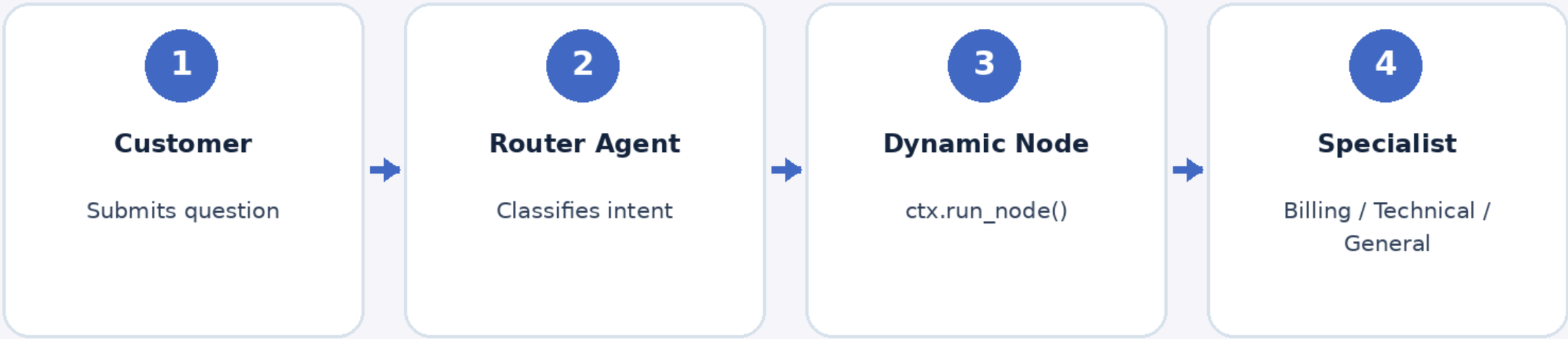
Batter and bowler branches invoke separate stats agents. Note: batter_stats_agent currently declares the bowler output schema, which should be corrected.

# Dynamic Routing

05 workflow_Graph

## Dynamic Graph Workflow — Support

**ROUTER NODE SCHEDULES A SPECIALIST AT RUNTIME**



# Collaborative Sub Agent

06 workflow_collaboration

## Multi-Agent Collaboration — Travel

**ONE COORDINATOR DELEGATES TO SPECIALISTS**

1

**Traveler**

Requests a trip plan

→

2

**Coordinator**

Delegates tasks

→

3

**Subagents**

Flights + Hotels + Activities

→

4

**Output**

Combined itinerary

# Human In the loop demo

07 workflow_humanintheloop

## Human-in-the-Loop — Refund

**REFUNDS OVER $100 REQUIRE A HUMAN DECISION**

1

**Customer**

Requests refund

2

**Refund Agent**

Calls approve_refund

3

**Decision**

≤ $100 auto; > $100 approval

4

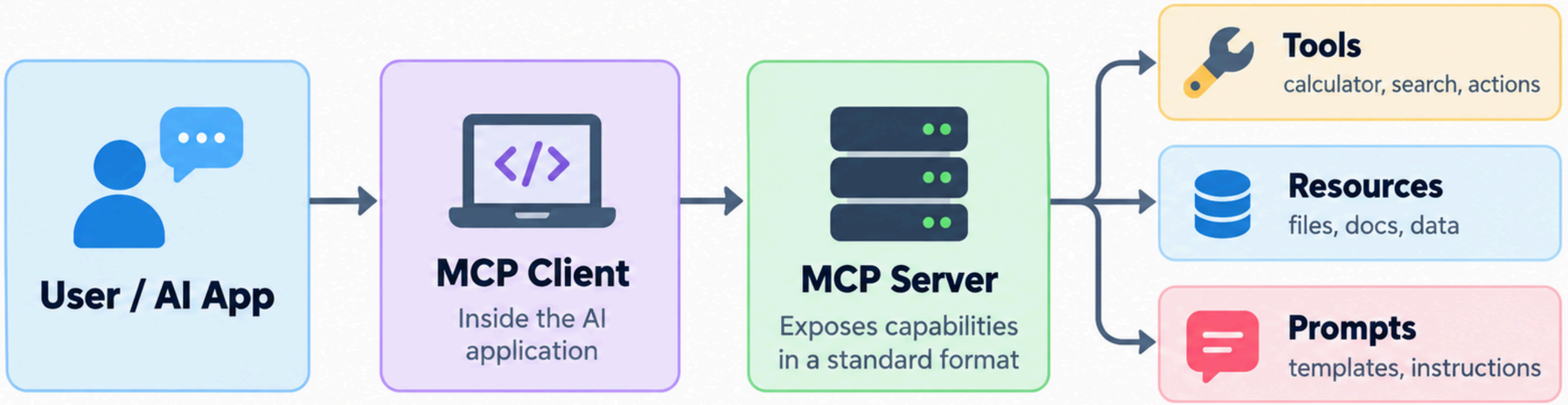
**Outcome**

Processed / Rejected / Waiting

# MCP Introduction

## Model Context Protocol (MCP)

A standard way for AI apps to connect to tools and data

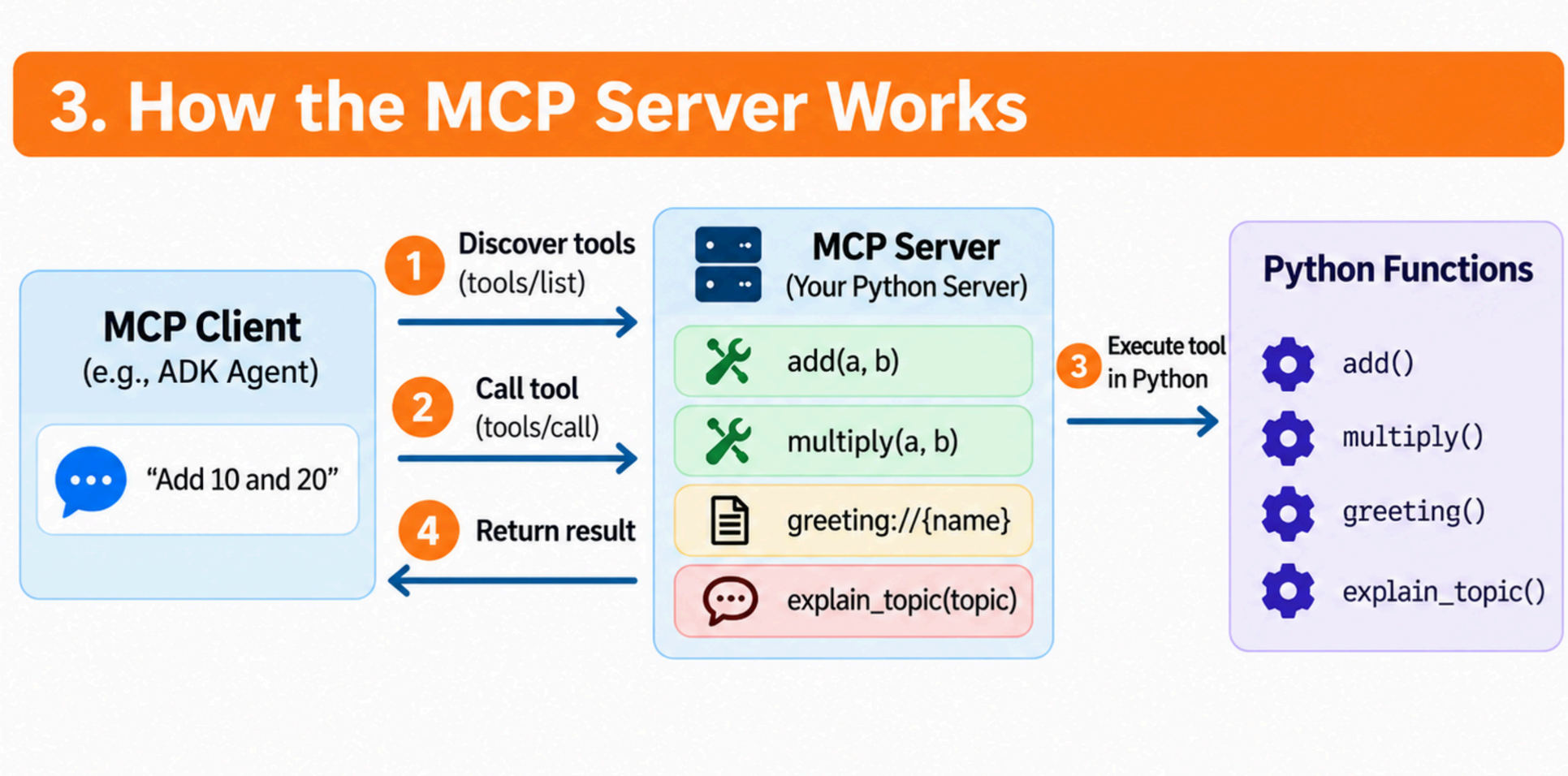



### How it works

1. AI app connects to MCP server
2. Server lists available tools/resources/prompts
3. AI chooses what it needs
4. Server returns results in a standard format

**MCP makes integrations reusable, standard, and easier to connect.**

# MCP Server and Client

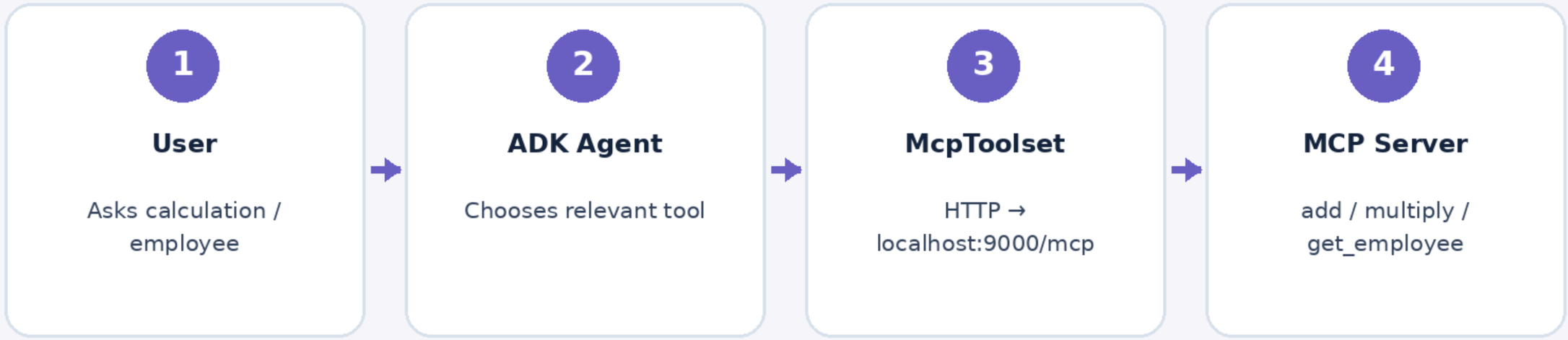


# MCP Interaction demo

08 mcpclient

## External MCP Server Integration

**ADK DISCOVERS TOOLS FROM A SEPARATE HTTP MCP SERVER**



# MCP Live demo - MCP Server

**MCP Python SDK v1.x API**

```
# Create a project directory
mkdir my-mcp-server
cd my-mcp-server

# Initialize a Python project
uv init

# Install the MCP SDK
uv add "mcp[cli]>=1.28,<2"
```

# MCP Live demo -  MCP Server

## MCP Python SDK v1.x API

uv run mcp dev '/Users/prasanna/mcpserver/server.py'

## MCP Inspector - to inspect the MCP

## Local MCP Server running in sandbox with access credentials

**Sandbox (MCP Apps): http://127.0.0.1:6275/sandbox**

**Auth token: 90bebe59d758b86d0ef69d8c7954708633797ac90ee4aad5c88fda755756ff20**

# MCP Live demo -  MCP Server

**MCP Python SDK v1.x API**

uv run mcp dev '/Users/prasanna/mcpserver/server.py'

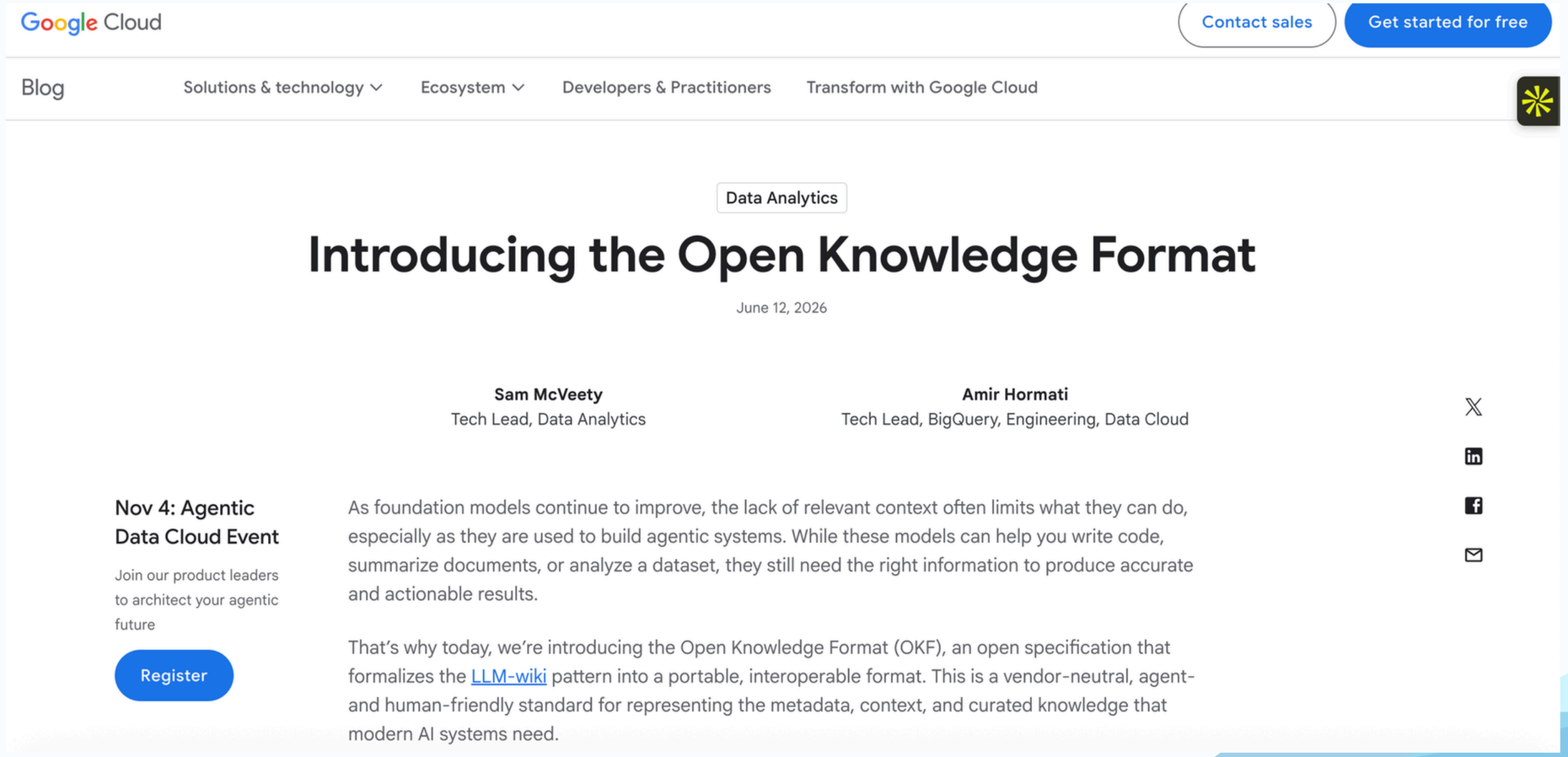
**MCP Inspector - to inspect the MCP**

# Short Term and Long Term memory Option for Agents from ADK

## Session service options

| Service | Storage | Survives Restart? |
| --- | --- | --- |
| InMemorySessionService | RAM | No |
| DatabaseSessionService | SQLite / Postgres | Yes |
| VertexAISessionsService | Google Cloud | Yes |

# Google Open Knowledge Format approach for Agent Knowledge Management.




https://github.com/GoogleCloudPlatform/open-knowledge-format

# Agents and Beyond

## Development

Eval

Gaurdrails

Multi Model Agent

A2A

Agent Security

Authentication and Authorization for Agents

## Deployment

Google Managed agent

Control Plane & Data Plane

Infra Security for Agents

Automated Agent creation and Deployment

## Post Deployment continues Monitoring

Agent Monitoring

Continues Evaluation & Finetuning

# Q&A

GET CONNECTED

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# THANK YOU